

Unconfoundedness II: Propensity Scores and Regression

BUSS975: Causal Inference in Financial Research

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Outline

1. Propensity scores
2. Regression
3. What matching cannot do
4. Sensitivity analysis
5. The NSW demonstration

Lecture 3a matched units cell by cell and paid for it twice — in bias and in variance.
This lecture replaces the whole vector X with a single number, and asks what that buys and what it costs.

1. Propensity Scores

Defining the propensity score

- **Propensity score:** $e(X) = \Pr(D = 1 \mid X)$
 - ▶ “the conditional probability of assignment to a particular treatment given a vector of observed covariates”
- **Balancing property** (Rosenbaum–Rubin 1983): $X \perp D \mid \Pr(D = 1 \mid X)$
 - ▶ $E[X \mid D = 1, \Pr(D = 1 \mid X)] = E[X \mid D = 0, \Pr(D = 1 \mid X)]$
 - ▶ Covariate distributions are similar between treatment and control conditional the propensity score.
- **Propensity score theorem** (Rosenbaum–Rubin 1983): if $(Y^1, Y^0) \perp\!\!\!\perp D \mid X$, then

$$(Y^1, Y^0) \perp\!\!\!\perp D \mid e(X)$$

- ▶ Conditioning on a **single number** does what conditioning on all of X does. Ten covariates collapse to one scalar — the curse of dimensionality defeated, for *conditioning*.

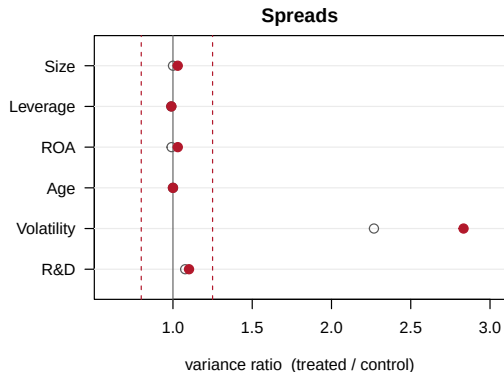
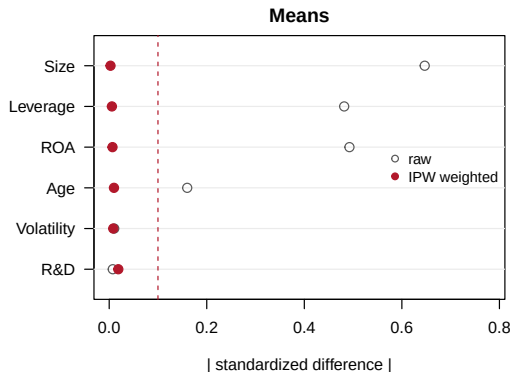
Checking balance: two numbers, not a t -test

The balancing property is testable. Report it covariate by covariate, before and after.

$$\text{SMD} = \frac{\bar{X}_1 - \bar{X}_0}{\sqrt{(s_1^2 + s_0^2)/2}}, \quad \text{VR} = \frac{s_1^2}{s_0^2}$$

- **Standardized mean difference.** Scale-free, so leverage and log assets are comparable. $|\text{SMD}| < 0.1$ is good; Rubin (2001) treats 0.25 as the outer limit past which regression adjustment is untrustworthy.
- Keep the denominator **fixed at its raw-sample value** when recomputing after weighting, or the yardstick moves with the thing you are measuring.
- **Variance ratio.** Means can match while spreads do not. Rubin: stay inside $[0.5, 2]$, ideally $[0.8, 1.25]$.
- **Do not use t -tests.** Balance is a property of *this sample*, not a population hypothesis — and p confounds imbalance with sample size, so pruning to a matched subsample raises it mechanically even as balance worsens (Imai, King and Stuart 2008).

A balance plot: means are not enough



Selection here is linear in Size, Leverage, ROA, Age — and **quadratic in Volatility**. The score was fitted with linear terms only.

Reading the balance plot

- **Left panel, the success.** Four covariates start badly imbalanced — Size at 0.65, Leverage and ROA near 0.49. After weighting, every one is under 0.02. This is the plot referees want, and it is the one most papers stop at.
- **Left panel, the trap.** Volatility looks *perfect* from the start: $SMD = -0.010$, and a t -test returns $p = 0.72$. On the evidence of means, nothing to see.
- **Right panel.** Its variance ratio is **2.27** — treated firms have more than twice the spread. Extreme volatility drives selection in *both* tails, so the means cancel and the distributions do not.
- **Weighting makes it worse**, $2.27 \rightarrow 2.83$, because the fitted score never contained vol^2 . Add that one term and the ratio falls to **1.03**.
- The lesson of this section: judge the score by **balance achieved, not by fit** — and balance means the whole distribution. Check squares and interactions, not just levels.

Estimating the score

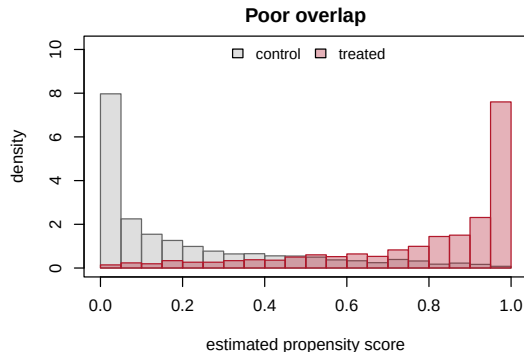
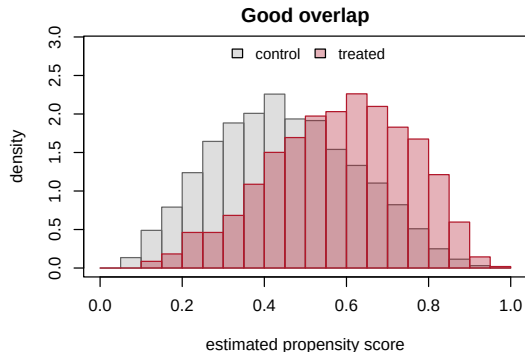
- You know the **true** score only when you ran the experiment yourself. Otherwise, it must be estimated: $\Pr(D_i=1 \mid X_i) = F(\beta X_i)$, fitted by **logit or probit**.
- Judge the specification by **balance achieved**, not by fit. A score with a high pseudo- R^2 but poor balance is a bad score; the reverse is fine.
- Fit, check balance, add interactions and polynomials where balance fails, refit. A legitimate specification search, because it never looks at Y — the whole stage is **blind to the outcome**, which protects against fishing. Say so in the paper.
- **The score creates no support that X lacks.** If a covariate cell is all-treated or all-control, collapsing X to a scalar hides that; it does not fix it.

Three things to do with the score

Once every unit has a $\hat{e}(X_i)$, there are three standard uses.

1. **Diagnose.** Plot $\hat{e}$ separately for treated and control units. This is the common-support check, and it costs nothing.
2. **Weight.** Reweight the sample to build a pseudo-experiment — inverse probability weighting.
3. **Match.** Pair treated to control units with similar scores, then difference — propensity score matching.

Visualizing overlap



Right panel: a treated mass at $\hat{e} \approx 1$ with no controls, and a control mass at $\hat{e} \approx 0$ with no treated. Those units have **no counterfactual**. Any estimate covering them is extrapolation, not comparison.

This plot belongs in every such paper

- Common support is verifiable, unlike unconfoundedness.
- What to do when it looks like the right panel:
 - ▶ **Trim** observations with extreme scores, or impose a caliper. Drop units with $\hat{e} \notin [0.1, 0.9]$. A rule of thumb, but a defensible and widely accepted one (Crump, Hotz, Imbens and Mitnik, 2009).
 - ▶ **Report the change in estimand.** After trimming you describe a subpopulation, not the original one. Say how many observations went and how the treated sample shifted.
 - ▶ Consider retreating to the *ATT*, which only needs overlap in one direction.
- What *not* to do: report the estimate and stay silent. The regression will happily extrapolate into a region with no data and give you a standard error that reflects none of that.

Inverse probability weighting

Rather than matching, **reweight**. Write $\hat{e}_i$ for $\hat{e}(X_i)$ and $\pi = \Pr(D=1)$.

$$ATE = E[Y_1 - Y_0] = E\left[\frac{D Y}{e(X)}\right] - E\left[\frac{(1-D) Y}{1-e(X)}\right]$$

$$ATT = E[Y_1 - Y_0 | D=1] = \frac{1}{\pi} \left(E[DY] - E\left[\frac{e(X)(1-D)Y}{1-e(X)}\right] \right)$$

$$\widehat{ATE}_{IPW} = \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i}{\hat{e}_i} - \frac{1-D_i}{1-\hat{e}_i} \right] Y_i$$

$$\widehat{ATT}_{IPW} = \frac{1}{n_T} \sum_{i=1}^n \left[D_i - (1-D_i) \frac{\hat{e}_i}{1-\hat{e}_i} \right] Y_i$$

- Both are **Horvitz–Thompson (1952)** forms.

Why *that* weight for the ATE?

We prove $E\left[\frac{D \cdot Y}{e(X)}\right] = E[Y^1]$. The $E[Y^0]$ half is symmetric.

$$E\left[\frac{D \cdot Y}{e(X)}\right] = E\left[\frac{D \cdot Y^1}{e(X)}\right]$$

switching equation (consistency): $D \cdot Y = D \cdot Y^1$

$$= E_X\left[E\left(\frac{D \cdot Y^1}{e(X)} \mid X\right)\right]$$

law of total expectation

$$= E_X\left[\frac{1}{e(X)} \cdot E(D \cdot Y^1 \mid X)\right]$$

$e(X)$ is fixed given X

$$= E_X\left[\frac{1}{e(X)} \cdot E(D \mid X) \cdot E(Y^1 \mid X)\right]$$

unconfoundedness (exchangeability): $D \perp\!\!\!\perp Y^1 \mid X$

$$= E_X\left[\frac{1}{e(X)} \cdot e(X) \cdot E(Y^1 \mid X)\right]$$

by definition, $E(D \mid X) = e(X)$

$$= E_X[E(Y^1 \mid X)] = E[Y^1]$$

cancel; law of total expectation

Why *that* weight for the ATT?

$$E[Y^1 \mid D=1] = E[Y \mid D=1] = \frac{E[DY]}{E[D]}.$$

$$E[Y^0 \mid D=1] = E_X \left[E(Y^0 \mid X, D=1) \mid D=1 \right]$$

law of total expectation, over $f(x \mid D=1)$

$$= E_X \left[E(Y^0 \mid X, D=0) \mid D=1 \right]$$

unconfoundedness — swap the arm

$$= E_X \left[E(Y^0 \mid X, D=0) \cdot \frac{e(X)}{\pi} \right]$$

Bayes: $f(x \mid D=1) = e(x)f(x)/\pi$

$$= E_X \left[\frac{E[(1-D)Y \mid X]}{1 - e(X)} \cdot \frac{e(X)}{\pi} \right]$$

$$E[(1-D)Y \mid X] = (1 - e(X))E[Y^0 \mid X, D=0]$$

$$= E \left[\frac{(1-D)Y e(X)}{(1 - e(X)) \pi} \right]$$

law of total expectation

Line 3: it converts an average over the *treated* covariate distribution into one over the *whole* population.

The intuition: the ATE rebuilds the population

Two kinds of firm. Small firms are **rarely** treated, large firms **usually** are.

Cell	N	e	treated	$\times 1/e$	controls	$\times 1/(1 - e)$
Small	500	0.2	100	$\times 5 = \mathbf{500}$	400	$\times 1.25 = \mathbf{500}$
Large	500	0.8	400	$\times 1.25 = \mathbf{500}$	100	$\times 5 = \mathbf{500}$
Total	1,000		500	1,000	500	1,000

- Only one small firm in five was treated, so that one must stand in for all five. Its weight is $1/0.2 = 5$.
- Every weighted entry comes back to 500: each arm is blown back up to the **full population**. That is the *pseudo-population*, and in it $D \perp\!\!\!\perp X$ by construction.
- Before weighting, small firms are 20% of the treated sample and 80% of the control sample. After weighting, 50% of both — the imbalance that caused the bias is gone.

... and the **ATT** reweights controls only

The *ATT* does not want the population. It wants the **treated** — and the treated are already a sample from that distribution, so they keep weight 1. Only the controls move.

Cell	controls	$\times e/(1 - e)$	=	treated
Small	400	0.25	100	100
Large	100	4.00	400	400

- **Why the odds?** In a cell with score e the sample splits $e : (1 - e)$ between the arms. Multiplying the controls by $e/(1 - e)$ makes their count match the treated count *exactly* — last column.
- The *ATE* weight asks “how many units like me are in the population?” The *ATT* weight asks “how many *treated* units like me are there?”
- Among large firms, 100 controls must cover 400 treated, so each carries weight 4. Push e toward 1 and that weight explodes: **this is where the ATT fails**, and why it needs overlap in one direction only, $e(X) < 1$, while the *ATE* needs $0 < e(X) < 1$.

Does it work?

Cell	e	$E[Y^0]$	$E[Y^1]$	effect	treated	controls
Small	0.2	10	12	2	100	400
Large	0.8	20	26	6	400	100

$$\widehat{E[Y^1]} = \frac{1}{1000} \left[\frac{100(12)}{0.2} + \frac{400(26)}{0.8} \right] = \frac{6000+13000}{1000} = 19$$

$$\widehat{E[Y^0]} = \frac{1}{1000} \left[\frac{400(10)}{0.8} + \frac{100(20)}{0.2} \right] = \frac{5000+10000}{1000} = 15$$

$$\widehat{ATT} = \underbrace{\frac{100(12)+400(26)}{500}}_{\text{treated mean} = 23.2} - \underbrace{\frac{400(10)(0.25)+100(20)(4)}{500}}_{\text{controls} \times \text{odds} = 18.0}$$

- Truth: $ATE = 4$, $ATT = 5.2$. The naive SDO is $23.2 - 12 = \mathbf{11.2}$.
- IPW returns $19 - 15 = \mathbf{4}$ and $23.2 - 18.0 = \mathbf{5.2}$. Both exactly right.
- $ATT > ATE$ because the effect is bigger for large firms, and large firms are where treatment concentrated. **Selection on gains — the two estimands correctly disagree.**

Why normalize the weights?

IPW is **sensitive to extreme propensity scores**: one observation with $\hat{e}$ near 0 or 1 gets an enormous weight and can dominate everything else.

Four treated firms with scores $\hat{e} = (0.5, 0.5, 0.5, 0.25)$, and **every one of them has** $Y = 10$. With $n = 8$:

$$\widehat{E}[Y^1] = \frac{1}{8} \left[\frac{10}{0.5} + \frac{10}{0.5} + \frac{10}{0.5} + \frac{10}{0.25} \right] = \frac{20 + 20 + 20 + 40}{8} = \frac{100}{8} = 12.5$$

- Every number in the data is 10. Any sensible estimator must return 10. This one returns 12.5.
- The reason: the weights are 2, 2, 2, 4 and they **sum to 10, not to 8**. Dividing by n assumes they sum to n — true in population, where $E[D/e(X)] = 1$, but never exactly true in a sample with $\hat{e}$ estimated.
- Fix: **divide by the weights you actually got, not by n** . Here $100/10 = 10$.

The normalized (Hájek) estimator

Hirano and Imbens (2001) normalize the weights **by their own sum within each group**, so the weights sum to one in each arm:

$$\widehat{ATE}_{\text{norm}} = \underbrace{\left[\sum_{i=1}^N \frac{Y_i D_i}{\hat{e}_i} \right] / \left[\sum_{i=1}^N \frac{D_i}{\hat{e}_i} \right]}_{\widehat{E[Y^1]}} - \underbrace{\left[\sum_{i=1}^N \frac{Y_i (1 - D_i)}{1 - \hat{e}_i} \right] / \left[\sum_{i=1}^N \frac{1 - D_i}{1 - \hat{e}_i} \right]}_{\widehat{E[Y^0]}}$$

- Each term is now a genuine **weighted average of observed outcomes**, so it must fall between the smallest and largest Y in that arm. The unnormalized version carries no such guarantee — which is exactly how it produced 12.5.
- Same repair for the *ATT*: divide the reweighted control sum by $\sum_i (1 - D_i) \frac{\hat{e}_i}{1 - \hat{e}_i}$ rather than by n_T .
- Default in Stata's `teffects ipw` and R's `WeightIt`.

Propensity score matching

- Match each treated unit to controls with a **similar score**, then difference.
- **Caliper** matching: refuse any match farther than some tolerance (say 0.001 on the score), *discarding* the treated unit if none qualifies. **Radius** matching: average all controls inside the tolerance.
- Two problems with both:
 - ▶ The tolerance is **chosen by the researcher** (Huntington-Klein et al. 2021).
 - ▶ Dropping treated units **changes the estimand** — the *ATT* summarised the *original* treated sample.
- The score is estimated, so its sampling error enters the standard errors (Abadie–Imbens 2016).
- King & Nielsen, 2019: Two units can share an identical score $e(X_i) = e(X_j)$ despite having completely opposite covariate profiles $X_i \neq X_j$. Matching purely on the scalar score can actually increase local covariate imbalance compared to multivariate distance matching (e.g., Coarsened Exact Matching).

What the score does and does not buy

- It buys **dimension reduction**, not identification. If X omits a confounder, $e(X)$ omits it too.
- Standard regression silently hides lack of common support through linear extrapolation, whereas plotting $\hat{e}(X)$ immediately exposes severe positivity violations.
- a high pseudo- R^2 in a treatment model means covariates perfectly predict treatment assignment ($e(X) \rightarrow 0$ or 1). This breaks the positivity assumption and causes extreme IPW variance. Conversely, a low pseudo- R^2 indicates that treatment assignment is nearly independent of X (resembling a randomized control trial).
- IPW shifts the modeling assumption from $E[Y \mid D, X]$ (outcome equation) to $P(D = 1 \mid X)$ (treatment assignment equation).

2. Regression

Regression as a conditioning strategy

- Regression belongs in this chapter. It is not a rival to matching — it is another way of executing the same substitution from Lecture 3a, §3, Identification Assumptions.
- Under unconfoundedness and **constant** treatment effects, $Y = \alpha + \beta D + X'\gamma + \varepsilon$ recovers β , because unconfoundedness delivers exactly the exogeneity condition OLS needs:

$$(Y^1, Y^0) \perp\!\!\!\perp D \mid X \implies E[\varepsilon \mid D, X] = 0$$

- Recall Lecture 1: the error term contains Y^0 . Selection bias was $\text{cov}(D, Y^0) \neq 0$. Conditioning on X is the attempt to make that covariance zero *within strata*.
- What regression adds that matching lacks: it **interpolates and extrapolates** with functional form, so empty cells do not stop it.
- What it hides: it does that silently.

Frisch–Waugh–Lovell

The coefficient on D in a multiple regression equals the coefficient from a **bivariate** regression of residualized Y on residualized D :

```
set.seed(975)
n <- 20000; X <- rnorm(n); D <- rbinom(n, 1, plogis(1.2 * X))
Y <- 2 * D + 3 * X + rnorm(n)
c(direct = unname(coef(lm(Y ~ D + X))[2]),
  fwl     = unname(coef(lm(resid(lm(Y ~ X)) ~ resid(lm(D ~ X))))[2]))
```

```
direct    fwl
2.018086 2.018086
```

- Identical to machine precision. “Controlling for X ” *means* “using only the variation in D that X cannot explain.”
- This is the cleanest statement of what a control variable does — and it tells you where the identifying variation comes from.
- It also warns you: if X explains nearly all of D , almost no variation remains, and the estimate rests on very few effective observations.

What FWL implies about weights

- FWL says OLS uses the residual variation in D . Where is that variation largest? Where treatment is **least predictable**.
- Formally, with X dummies, OLS weights each stratum by $N_j p_j(1 - p_j)$ — the **conditional variance of treatment** within it.
- So OLS gives most weight to strata where treatment is closest to a coin flip, and least where it is nearly deterministic.
- That is a defensible **efficiency** motive: those strata carry the most information per observation.
- It also means **OLS is not estimating the ATT** — or the ATE — when effects are heterogeneous. It estimates a variance-weighted average with no policy interpretation.

Regression adjustment (imputation)

- A more flexible use of regression that mirrors matching directly:
 1. Fit $\hat{\mu}^1(x)$ on the **treated** and $\hat{\mu}^0(x)$ on the **controls**, separately.
 2. For every unit, impute both potential outcomes and average the difference:

$$\widehat{ATE} = \frac{1}{n} \sum_i (\hat{\mu}^1(X_i) - \hat{\mu}^0(X_i))$$

- Separate models allow the covariates to matter differently under treatment and control — exactly what a single β on D forbids.
- This is the same imputation logic as matching, with a model in place of a neighbour.
- Combining it with weighting gives **doubly robust** estimators: consistent if *either* the outcome model or the propensity model is right. A genuinely useful hedge, and increasingly the default in applied work.

Matching versus regression

- Both rely on the **same** identifying assumption: unconfoundedness. Neither creates exogenous variation.
- The difference is in *how they weight strata* when the effect is heterogeneous.
 - ▶ **Matching (ATT)** weights stratum j by the number of **treated** units in it.
 - ▶ **OLS with X dummies** weights stratum j by $N_j p_j(1 - p_j)$.
- They also differ in what they do about missing cells: matching refuses; regression extrapolates.
- So when the two disagree, ask two questions in order: *is it the weights, or is it the extrapolation?* Both are diagnosable.

The same example under both

Two strata. $p_{\text{small}} = 0.5$, $p_{\text{large}} = 0.1$; effects +2 and +3; 100 firms per stratum.

	Weight on small	Weight on large	Estimate
Matching (<i>ATT</i>): treated counts	$50/60 = 0.833$	$10/60 = 0.167$	2.167
OLS: $N_j p_j (1 - p_j)$	$25/34 = 0.735$	$9/34 = 0.265$	2.265
<i>ATE</i> : population shares	0.5	0.5	2.500

```
X <- c(rep(0, 100), rep(1, 100))
D <- c(rep(c(1, 0), each = 50), rep(c(1, 0), times = c(10, 90)))
Y <- ifelse(X == 0, 0, 5) + ifelse(X == 0, 2, 3) * D
round(unnname(coef(lm(Y ~ D + factor(X)))[2]), 3)
```

[1] 2.265

Reading the comparison

- The regression returns 2.265 — exactly the variance-weighted average, not the *ATT* and not the *ATE*.
- The gap is small here. Angrist and Pischke argue it usually is, in practice.
- But “usually small” is not “zero”, and the difference is entirely about **which firms your estimate describes**:
 - ▶ Matching describes the treated.
 - ▶ OLS describes wherever treatment was most unpredictable.
- **When it matters most**: when treatment probability varies a lot across strata *and* effects are heterogeneous. Both are typical in finance.
- This is the same variance-weighting issue that resurfaces, far more destructively, in staggered difference-in-differences.

3. What Matching Cannot Do

Matching creates no new variation

- Matching **does not introduce exogenous variation**. It reorganizes and reweights the data you already have, using observables you already measured.
- Consequence: if your OLS regression was biased by endogeneity, **the matched estimate is biased too**.
- Contrast this with the designs still to come:
 - ▶ **IV** imports variation from outside the model.
 - ▶ **RDD** exploits assignment that is locally as good as random.
 - ▶ **DiD** differences out fixed unobservables.
- Matching belongs to a different category. It is a *conditioning strategy*, like regression — not a research design.

What it cannot fix

Matching is powerless against:

- **Unobserved confounders.** You cannot match on what you cannot measure.
- **Reverse causality.** Matching on X says nothing about whether Y caused D .
- **Measurement error** in treatment or outcome.
- **Bad controls.** Matching on a post-treatment variable is exactly as damaging as controlling for it — Lecture 2 applies unchanged, and the collider F of Lecture 3a, §2, Confounders, Covariates, and Colliders, is the warning.

You cannot match on what you cannot measure.

So what is it good for?

Given all that, matching still earns its place:

1. **A non-parametric benchmark.** It estimates the effect without assuming a functional form. If matching and OLS disagree sharply, your specification is doing more work than you thought.
 2. **Common-support discipline.** Matching refuses to compare units with no counterpart. Regression silently extrapolates into regions with no data.
 3. **Diagnostic value.** The propensity score distribution reveals overlap problems that a regression table hides completely.
 4. **Robustness when the functional form is wrong.** Matching can outperform a misspecified regression in finite samples.
- Use it as a *complement* to regression and a check on it — not as an identification strategy.

4. Sensitivity Analysis

The assumption you cannot test, revisited

- Two things this course has said are both true and both final: unconfoundedness is **untestable**, and no amount of matching creates variation that fixes an unobserved confounder.
- Neither is the end of the conversation. **Sensitivity analysis** changes the question:

Not “is my design valid?” — unanswerable. But:

how large would a hidden confounder have to be to overturn my conclusion?

- The output is not a p -value. It is a **threshold**, plus an argument about whether a confounder that large is plausible in your setting. The argument is yours; the threshold is arithmetic.
- Two families, matching the two routes of §1, Propensity Scores, and §2, Regression:
 - ▶ **Design-based**, for matched pairs: Rosenbaum bounds.
 - ▶ **Regression-based**, for the coefficient: Oster’s δ , and the Cinelli–Hazlett robustness value.
- Referees in finance now ask for one of these by name. Budget a table for it.

Rosenbaum bounds: one parameter, Γ

- Matching made two units identical in X . Unconfoundedness then says they had **equal** probability of treatment. Suppose they did not.
- Let $\Gamma \geq 1$ bound how unequal. For any two units i, j with $X_i = X_j$,

$$\frac{1}{\Gamma} \leq \frac{\pi_i(1 - \pi_j)}{\pi_j(1 - \pi_i)} \leq \Gamma.$$

- $\Gamma = 1$ **is** unconfoundedness. $\Gamma = 2$ says a hidden variable could leave one unit of the pair twice as likely, in odds, to be treated.
- For each Γ , compute the **worst case** over every hidden assignment consistent with that Γ , and report the resulting bound on the p -value of the randomization test.
- Then read off Γ^* : the point where the upper bound crosses your threshold.
- Note what Γ^* is *not*. It is a statement about how much hidden bias your **design** can absorb. It never says the design is fine.

Reading a Rosenbaum bound

200 matched pairs; one-sided Wilcoxon signed-rank test on the within-pair differences.

Γ	1.00	1.10	1.20	1.30	1.40	1.50	1.75	2.00
upper bound on p	0.0001	0.0008	0.0040	0.0149	0.0415	0.0922	0.3354	0.6386

- At $\Gamma = 1$ the result looks bulletproof: $p = 0.0001$. It is not. $\Gamma^* = 1.42$.
- A hidden variable that shifts the odds of treatment by 42% between two otherwise identical units is enough to make this finding insignificant.
- This is the standard and slightly deflating experience. Strong p -values routinely coexist with small Γ^* , because the two answer different questions.
- Which leaves the question the table cannot answer: **is 1.42 big?**

Calibrating Γ : what does 1.42 mean?

- Γ is one number doing the work of two. Rosenbaum and Silber (2009) split it back apart: let Λ be the odds ratio linking the hidden variable to **treatment**, and Δ the odds ratio linking it to the **outcome**. Then

$$\Gamma = \frac{\Lambda\Delta + 1}{\Lambda + \Delta}.$$

- Every (Λ, Δ) on that curve produces the same Γ . At $\Gamma^* = 1.42$:

Λ (odds of treatment)	2.43	3.00	5.00
Δ (odds of a higher outcome)	2.43	2.07	1.71

- Now the question is one a *finance* audience can argue about: is there a plausible omitted characteristic that roughly **doubles** the odds of treatment and **doubles** the odds of a better outcome? Often the honest answer is yes.
- Benchmark it.** Compute Λ and Δ for your strongest *measured* confounder. If the killer confounder must be twice as strong as anything you measured, that is an argument. If half as strong will do, so is that.
- The bound is worst-case, so Γ^* understates the true robustness.

Oster's δ : coefficient stability, done properly

- The informal version is in every seminar: *"the coefficient barely moved when I added controls, so unobservables cannot matter much."*
- That intuition is not wrong, it is **unfinished**. It ignores how much the controls explained. A coefficient that is stable while R^2 barely moves has demonstrated nothing.
- Altonji, Elder and Taber (2005) turned it into an argument; Oster (2019) turned it into an estimator. Two inputs:
 - ▶ δ — selection on unobservables **relative to** selection on observables. $\delta = 1$ means they are equally severe.
 - ▶ $R_{\max}$ — the R^2 of the infeasible regression that included the unobservables.
- With $(\hat{\beta}, \hat{R})$ from the short regression and $(\tilde{\beta}, \tilde{R})$ from the controlled one,

$$\beta^* \approx \tilde{\beta} - \delta (\hat{\beta} - \tilde{\beta}) \frac{R_{\max} - \tilde{R}}{\tilde{R} - \hat{R}}.$$

- Two ways to report it: fix $\delta = 1$ and give β^* ; or set $\beta^* = 0$ and solve for the δ that would be needed. Oster's calibration from randomized trials: $\delta = 1$, $R_{\max} = \min(1.3 \tilde{R}, 1)$.

Oster in practice: a warning

Simulated data, one unobserved confounder, **true effect** = 1.000:

	β	R^2
short: $Y \sim D$	2.086	0.452
controlled: $Y \sim D + X$	1.743	0.484
infeasible: $Y \sim D + X + U$	1.014	0.562

- Oster's default rule sets $R_{\max} = 1.3 \times 0.484 = 0.629$, and returns $\beta^* = \mathbf{0.201}$ — a correction that overshoots the truth by a mile.
- Feed it the *true* $R_{\max} = 0.562$ and it returns $\beta^* = \mathbf{0.918}$, almost exactly right. The method works. The **guess** was what failed.
- Meanwhile the δ test passes: the δ that would drive β^* to zero is $1.13 > 1$.
- **The lesson.** Report δ , which is a bounded comparison you can defend. Do not report β^* as if it were a point estimate — it is a linear function of a number you guessed. If you show it, show it across a range of $R_{\max}$.

Cinelli and Hazlett (2020): the robustness value

- Same problem, third parameterization — and the one that travels best, because it uses **partial** R^2 rather than odds ratios or selection ratios.
- RV_q is the minimum share of the residual variation that a confounder must explain in **both** D and Y in order to move the estimate by $100q\%$.
- In the regression above: $t = 47.1$ and $RV_{q=1} = \mathbf{0.48}$. Any confounder explaining less than 48% of what is left in both D and Y cannot drive the effect to zero.
- Is that reassuring? Check it against the confounder we planted: U explains 15% of the residual variation in Y and 34% in D . Below the bar — and indeed it does not kill the effect, it moves it from 1.743 to 1.014.
- The feature that sells it: **benchmark bounds**. Express RV as a multiple of an observed covariate's partial R^2 — “the confounder would have to be $3\times$ as important as firm size” is a sentence a referee can evaluate.
- Software: `sensemkr` (R and Stata), `psacalc` for Oster, `rbounds` and `mhbounds` for Rosenbaum.

5. The NSW Demonstration

The 1970s empirical crisis

- The methods in this lecture were not adopted because they were elegant. They were adopted after a crisis of confidence in observational work.
- Job-training programs in the 1970s were evaluated observationally, and the estimates were all over the place — depending, apparently, on the analyst.
- The **National Supported Work (NSW) Demonstration** was different: it *randomized* participation. So the experimental effect was known.
- That created something rare and valuable: a **benchmark**. You could now ask whether observational methods, applied to the same treated units, recover the experimental answer.
- Everything in this lecture gets graded against it.

What LaLonde did

- LaLonde (1986) ran the decisive test. He took the NSW experiment, **threw away the experimental control group**, and replaced it with comparison groups built from survey data (CPS and PSID).
- He then applied the standard observational toolkit of the day and compared the results to the experimental benchmark of roughly **\$1,800** in annual earnings gains.
- The observational estimates ranged from a fraction of the benchmark to **the wrong sign** — and which one you got depended heavily on how the comparison group was built.
- The paper was devastating and hugely influential. It is a major reason the profession moved toward IV, RDD, and DiD — the other river.

Dehejia and Wahba: not so fast

- **Dehejia and Wahba (1999, 2002)** revisited LaLonde's data with propensity score methods.
- Their finding: the observational estimates *did* recover the experimental benchmark reasonably well — **once common support was imposed**.
- The key diagnosis: much of LaLonde's failure was an **overlap** problem, not only a confounding problem. The CPS and PSID comparison groups contained enormous numbers of people utterly unlike the NSW participants.
- Discard the non-comparable units and the methods perform far better.
- This is why the distinction of Lecture 3a, §3, Identification Assumptions, matters so much. Two different assumptions were failing, and only one of them is checkable.

Smith and Todd: not so fast either

- **Smith and Todd (2005)** re-examined the Dehejia–Wahba result and found it fragile.
- The success was **sensitive to the subsample** analysed and to the **propensity score specification** chosen. Reasonable alternative choices undid it.
- Their charge is uncomfortable and worth stating plainly: the specification that worked may have been found *because* the experimental answer was already known.
- You cannot run that search in real research. In a genuine application there is no benchmark to check against — that is the entire point of doing the study.

The honest summary

- Selection on observables **can** work — when overlap is genuine and the confounders are truly measured.
- Neither condition is verifiable in the setting where you need them:
 - ▶ Unconfoundedness is **untestable**, always.
 - ▶ Common support is **checkable**.
- So the practical lesson is asymmetric: do the check you can do, and argue explicitly for the assumption you cannot.
- **A useful exercise:** the NSW data are freely available. Apply every method in this lecture and compare them to the \$1,800 benchmark. Cunningham's chapter walks through it, and it is the fastest way to develop instincts about when these methods hold up.

Wrapping Up

Summary [Part 1]: the score and what it does

- **The propensity score** reduces X to one number and defeats the curse of dimensionality for *conditioning*. It does nothing for **identification** — unconfoundedness is still assumed, never delivered.
- **Judge the score by balance achieved, not by fit.** A high pseudo- R^2 with bad balance is a bad score; the reverse is fine. The whole stage is blind to Y , which is what makes the specification search legitimate.
- **Balance means the distribution, not the mean.** Standardized differences *and* variance ratios. Our Volatility covariate had $SMD = -0.010$ and a variance ratio of 2.27.
- **IPW** reweights the sample into a pseudo-experiment: $1/\hat{e}$ for treated and $1/(1 - \hat{e})$ for controls builds the population; $\hat{e}/(1 - \hat{e})$ on controls alone rebuilds the *treated* distribution. Watch the tails, and normalize.
- **Propensity score matching** is the one use with a serious case against it: King and Nielsen (2019) show it targets complete randomization when you could have had blocking, so pruning can *increase* imbalance.

Summary [Part 2]: the verdict

- **Regression is a conditioning strategy too**, not a rival family. FWL shows it uses the residual variation in D , which is why it weights strata by $N_j p_j (1 - p_j)$: 2.167 versus matching's 2.265 in our example.
- **Matching creates no new exogenous variation.** If OLS was endogenous, matching is endogenous. It cannot fix an unobserved confounder, reverse causality, or a bad control.
- **Sensitivity analysis is the honest response** to an untestable assumption. Report Γ^* , or δ , or a robustness value — and calibrate it against a covariate you actually measured.
- **NSW**: these methods can work when overlap is real and the confounders are measured — and you can only ever verify one of those two.
- Next lecture: designs that *create* variation instead of assuming it away.

A checklist for your own paper

1. State the **estimand** before the estimator. *ATE*, *ATT*, or *ATU* — and say why the policy question needs that one.
2. Defend the **assignment mechanism** in institutional terms. Name the confounders and say how you measured them.
3. Justify every covariate as **pre-treatment**. Any post-treatment control needs an argument, not a footnote.
4. **Plot the overlap**. Report what you trimmed and how the estimand changed.
5. Report **balance**, not fit: standardized differences and variance ratios, before and after.
6. Report **matching and regression**. If they disagree, diagnose whether it is weights or extrapolation.
7. Run a **sensitivity analysis**, and calibrate the threshold against an observed covariate.
8. Run a **placebo** on a pre-treatment period.
9. Say plainly what would break the design — and what unobservable would have to exist.

References

- Cunningham, S. *Causal Inference: The Remix*, Ch. 5.
- Rosenbaum, P. and D. Rubin (1983). The Central Role of the Propensity Score in Observational Studies for Causal Effects. *Biometrika* 70, 41–55.
- Hirano, K. and G. Imbens (2001). Estimation of Causal Effects Using Propensity Score Weighting. *Health Services and Outcomes Research Methodology* 2, 259–278.
- Imbens, G. (2004). Nonparametric Estimation of Average Treatment Effects under Exogeneity. *Review of Economics and Statistics* 86, 4–29.
- King, G. and R. Nielsen (2019). Why Propensity Scores Should Not Be Used for Matching. *Political Analysis* 27, 435–454.
- Ben-Michael, E., A. Feller and J. Rothstein (2021). The Augmented Synthetic Control Method. *JASA* 116, 1789–1803.
- LaLonde, R. (1986). Evaluating the Econometric Evaluations of Training Programs with Experimental Data. *American Economic Review* 76, 604–620.
- Dehejia, R. and S. Wahba (1999). Causal Effects in Nonexperimental Studies. *JASA* 94, 1053–1062.
- Smith, J. and P. Todd (2005). Does Matching Overcome LaLonde's Critique of Nonexperimental Estimators? *Journal of Econometrics* 125, 305–353.

References: sensitivity analysis

- Rosenbaum, P. (2002). *Observational Studies*, 2nd ed. Springer.
- Rosenbaum, P. and J. Silber (2009). Amplification of Sensitivity Analysis in Matched Observational Studies. *JASA* 104, 1398–1405.
- Altonji, J., T. Elder and C. Taber (2005). Selection on Observed and Unobserved Variables: Assessing the Effectiveness of Catholic Schools. *Journal of Political Economy* 113, 151–184.
- Oster, E. (2019). Unobservable Selection and Coefficient Stability: Theory and Evidence. *Journal of Business & Economic Statistics* 37, 187–204.
- Cinelli, C. and C. Hazlett (2020). Making Sense of Sensitivity: Extending Omitted Variable Bias. *Journal of the Royal Statistical Society Series B* 82, 39–67.